

Mock Midterm — Form A — answer key

Time allowed: 120 minutes · 100 points

QUICK KEY — MULTIPLE CHOICE

Part I: 1-B 2-B 3-C 4-B 5-B 6-B 7-A 8-D 9-A 10-B 11-C 12-C 13-B 14-B 15-A 16-C 17-C 18-D 19-D 20-C 21-D
22-C 23-A 24-A 25-D 26-A 27-B 28-C 29-D 30-D

Part I — Multiple choice

30 questions × 1 pt = 30 pts

1. MULTIPLE CHOICE

A noiseless channel has a bandwidth of 3 kHz and uses 4 signal levels. What is the maximum bit rate?

- A. 6 kbps
- B. 12 kbps ✓ correct
- C. 3 kbps
- D. 24 kbps

Why: Nyquist: bit rate = $2 \times \text{bandwidth} \times \log_2 L = 2 \times 3,000 \times \log_2 4 = 2 \times 3,000 \times 2 = 12,000$ bps. L02 p29; L02 p30

2. MULTIPLE CHOICE

According to the slides, why is analog transmission needed?

- A. Analog signals always travel faster than digital signals
- B. Bandwidth is limited, so the digital signal is modulated into an analog one ✓ correct
- C. Digital signals cannot travel over copper or fiber
- D. Analog transmission needs no modulator or demodulator

Why: Slide 17: analog transmission is needed when bandwidth is limited; digital signals are modulated into analog signals and transmitted. L01 p17

3. MULTIPLE CHOICE

What is **baseband** transmission?

- A. Changing a digital signal into an analog signal before sending it
- B. Sending analog signals over a bandpass channel
- C. Sending a digital signal over a channel without changing it into an analog signal ✓ correct
- D. Sending a signal only over wireless media

Why: Baseband sends the digital signal as it is, over a wide-bandwidth (dedicated) medium. Changing it into an analog signal by modulation, which lets us use a bandpass channel, is broadband transmission. L02 p16; L02 p18

4. MULTIPLE CHOICE

Which digital-to-analog scheme varies **only the amplitude** of the carrier?

- A. FSK
- B. ASK ✓ correct
- C. PSK
- D. QAM

Why: ASK changes the amplitude; FSK changes frequency, PSK changes phase, and QAM changes amplitude and phase together. L04 p8

5. MULTIPLE CHOICE

A sine wave starts at time 0 at its highest point and then falls. In the slide's convention its phase is:

- A. 0°
- B. 90° ✓ correct
- C. 180°
- D. 270°

Why: Slide 8: 0° starts at 0 and rises; 90° starts at the peak; 180° starts at 0 and falls; 270° starts at the trough. L02 p8

6. MULTIPLE CHOICE

Which is the correct order of the three PCM steps?

- A. Quantization → sampling → encoding
- B. Sampling → quantization → binary encoding ✓ correct
- C. Encoding → sampling → quantization
- D. Sampling → encoding → quantization

Why: PCM samples the analog signal (PAM), quantizes each sample to a zone midpoint, then encodes the zone number in binary. L03 p62

7. MULTIPLE CHOICE

Binary ASK with $d = 1$ must fit in an available bandwidth of 40 kHz. What is the maximum bit rate?

- A. 20 kbps ✓ correct
- B. 10 kbps
- C. 40 kbps
- D. 80 kbps

Why: $B = (1 + d) \times S \rightarrow 40 \text{ kHz} = 2S \rightarrow S = 20 \text{ kbaud}$. Binary ASK has $r = 1$, so $N = 20 \text{ kbps}$. L04 p12

8. MULTIPLE CHOICE

The power of a signal is 2 mW and the power of the noise is 2 μW . What is SNR_{dB} ?

- A. 3 dB
- B. 20 dB
- C. 25 dB
- D. 30 dB ✓ correct

Why: $\text{SNR} = 2 \text{ mW} / 2 \mu\text{W} = 2,000 \mu\text{W} / 2 \mu\text{W} = 1,000$, so $\text{SNR}_{\text{dB}} = 10 \log_{10} 1,000 = 30 \text{ dB}$. Convert both powers to the same unit first. L02 p25; L02 p26

9. MULTIPLE CHOICE

A composite signal is made of sine waves with frequencies f , $3f$ and $9f$, where $f = 200 \text{ Hz}$. What is its bandwidth?

- A. 1,600 Hz ✓ correct
- B. 1,800 Hz
- C. 200 Hz
- D. 600 Hz

Why: The components are 200, 600 and 1,800 Hz. Bandwidth = highest – lowest = $1,800 - 200 = 1,600 \text{ Hz}$ ($= 8f$). 1,800 Hz is only the highest frequency. L02 p12; L02 p13

10. MULTIPLE CHOICE

In asynchronous transmission, each byte is framed by:

- A. a start bit 1 and a stop bit 0
- B. a start bit 0 and one or more stop bits 1 ✓ correct
- C. a flag byte at each end
- D. no extra bits at all

Why: One start bit (0) begins every byte and one or more stop bits (1) end it; gaps may follow. L03 p92

11. MULTIPLE CHOICE

In an mBnL code, $2^m \leq L^n$ must hold. Which of these codes is **not** valid?

- A. 2B1Q
- B. 8B6T
- C. 3B1Q ✓ correct
- D. 4B3T

Why: 3B1Q needs $2^3 = 8$ data patterns but only $4^1 = 4$ signal patterns exist. The others pass: 2B1Q $4 \leq 4$, 8B6T $256 \leq 729$, 4B3T $16 \leq 27$ (slide 36). L03 p35; L03 p36

12. MULTIPLE CHOICE

HDB3 substitutes four zeros with **000V** when the number of nonzero pulses since the last substitution is odd. What does it use when that number is **even**?

- A. 000V
- B. 000VB0VB
- C. B00V ✓ correct
- D. 0000 (no substitution)

Why: Slide 60: even → B00V, odd → 000V — in both cases the total number of nonzero pulses becomes even. 000VB0VB is the B8ZS pattern for eight zeros. L03 p60

13. MULTIPLE CHOICE

What is **baseline wandering**?

- A. A slow change in the sender's clock frequency
- B. A drift of the receiver's running-average baseline, caused by a long string of 0s or 1s ✓ correct
- C. The delay between the first and the last bit of a block
- D. Noise that the channel adds to the signal

Why: In decoding, the receiver keeps a running average of the received signal power — the baseline. A long run of the same level drags that average away, so the receiver has trouble telling 0s from 1s (slide 11). L03 p11

14. MULTIPLE CHOICE

Which OSI layer frames the data and moves frames error-free between two end nodes over the physical layer?

- A. Physical
- B. Data link ✓ correct
- C. Network
- D. Transport

Why: The data link layer formats data into frames and enables error-free frame transfer between two end nodes over the physical layer. L01 p4; L01 p5

15. MULTIPLE CHOICE

A system uses NRZ-I to send data at 2 Mbps. What is the minimum bandwidth (average case)?

- A. 1 MHz ✓ correct
- B. 500 kHz
- C. 2 MHz
- D. 4 MHz

Why: NRZ-I has $r = 1$ and the average case uses $c = 1/2$: $S = c \times N \times 1/r = 1/2 \times 2 \text{ Mbps} \times 1 = 1 \text{ Mbaud}$. The minimum bandwidth equals the signal rate, $B_{\min} = S = 1 \text{ MHz}$ (Example 3 on slide 25 does the same with 1 Mbps). L03 p25

16. MULTIPLE CHOICE

According to the slides, a LAN typically connects computers within:

- A. 50 m
- B. 500 m
- C. 5 km ✓ correct
- D. 50 km

Why: Slides 6–7: a LAN is a connection of computers typically within 5 km. L01 p6; L01 p7

17. MULTIPLE CHOICE

Which IEEE standard defines wireless media access control and physical layer specifications (the Wi-Fi family)?

- A. 802.3
- B. 802.5
- C. 802.11 ✓ correct
- D. 802.16

Why: 802.11 = wireless MAC and PHY. 802.3 is Ethernet (CSMA/CD), 802.5 token ring, 802.16 wireless local loop. L01 p9; L01 p12

18. MULTIPLE CHOICE

Data at 8 Mbps is block-coded with 4B/5B before line coding. What is the bit rate on the line after block coding?

- A. 6.4 Mbps
- B. 8 Mbps
- C. 16 Mbps
- D. 10 Mbps ✓ correct

Why: 4B/5B replaces every 4 bits by 5, so $N' = N \times 5/4 = 8 \times 5/4 = 10$ Mbps (slide 54: 1 Mbps becomes 1.25 Mbps). 6.4 Mbps would be $N \times 4/5$ — the wrong direction. L03 p48; L03 p54

19. MULTIPLE CHOICE

A periodic signal has a period of 4 ms. What is its frequency?

- A. 4 Hz
- B. 4 kHz
- C. 25 Hz
- D. 250 Hz ✓ correct

Why: Frequency and period are reciprocals: $f = 1/T = 1 / 0.004 \text{ s} = 250 \text{ Hz}$. (Slide 7: a period of 1/4 s means 4 Hz.) L02 p7

20. MULTIPLE CHOICE

Which statement about signals is correct?

- A. Only analog signals can be periodic
- B. A nonperiodic signal repeats its pattern in every period
- C. Both analog and digital signals can be periodic or nonperiodic ✓ correct
- D. Digital signals can never be periodic

Why: A periodic signal repeats a pattern over identical periods; a nonperiodic signal shows no pattern that repeats over time. Both analog and digital signals can be either kind (in practice we commonly use periodic analog and nonperiodic digital signals). L02 p5

21. MULTIPLE CHOICE

Why is PSK more robust than ASK?

- A. PSK uses a higher carrier frequency
- B. PSK needs no carrier
- C. PSK has a lower baud rate
- D. Noise mostly alters amplitude, and PSK keeps the amplitude constant ✓ correct

Why: The slide: "PSK is much more robust than ASK as it is not that vulnerable to noise, which changes amplitude." L04 p21

22. MULTIPLE CHOICE

Using the slides' QPSK mapping (first bit → I carrier), which dibit is sent at phase 135°?

- A. 00
- B. 11
- C. 01 ✓ correct
- D. 10

Why: 135° is the upper-left point: I negative (first bit 0), Q positive (second bit 1) → 01. L04 p29

23. MULTIPLE CHOICE

Which description of **AMI** matches the slide notes?

- A. No DC component or baseline wandering, but no self-synchronization during a long run of 0s ✓ **correct**
- B. A DC component, but good self-synchronization
- C. Self-synchronization and error detection, but a DC component
- D. No DC component and a transition in the middle of every bit

Why: Slide 33: AMI has no DC component or baseline wandering, but a long run of 0s gives no signal transitions, so it cannot self-synchronize; it has no error detection. (The summary table on slide 46 marks AMI "DC", which contradicts slide 33.) L03 p33

24. MULTIPLE CHOICE

A signal travels through a cable and its power is reduced to one-half. The change is about:

- A. -3 dB ✓ **correct**
- B. -6 dB
- C. +3 dB
- D. -0.5 dB

Why: $\text{dB} = 10 \log_{10}(P_2/P_1) = 10 \log_{10}(0.5) = -3 \text{ dB} (-3.01)$. The decibel is negative when the signal is attenuated. L02 p21; L02 p22

25. MULTIPLE CHOICE

A low-pass signal contains frequencies from 0 to 5 kHz. What is the minimum sampling rate?

- A. 5000 samples/s
- B. 2500 samples/s
- C. 20,000 samples/s
- D. 10,000 samples/s ✓ **correct**

Why: Nyquist: $f_s \geq 2 \times f_{\text{max}} = 2 \times 5 \text{ kHz} = 10,000 \text{ samples/s}$. L03 p66; L03 p70

26. MULTIPLE CHOICE

A signal is quantized into $L = 32$ levels and sampled at 8000 samples/s. What is the bit rate?

- A. 40 kbps ✓ **correct**
- B. 256 kbps
- C. 32 kbps
- D. 64 kbps

Why: $n_b = \log_2 32 = 5$ bits per sample, so $N = 5 \times 8000 = 40 \text{ kbps}$. L03 p75; L03 p79

27. MULTIPLE CHOICE

Which line code has a level transition in the middle of **every** bit and uses only **two** voltage levels?

- A. RZ
- B. Manchester ✓ **correct**
- C. AMI
- D. NRZ-I

Why: Manchester (and differential Manchester) always change level in the middle of the bit using only +V and -V (slide 28). RZ also has a mid-bit transition but needs three levels (+, 0, -); AMI has three levels and no mid-bit transition; NRZ-I only changes level for a 1. L03 p26; L03 p28

28. MULTIPLE CHOICE

An analog signal carries 3 bits per signal element at 2000 baud. What is the bit rate?

- A. 667 bps
- B. 2000 bps
- C. 6000 bps ✓ **correct**
- D. 8000 bps

Why: $S = N \times 1/r$, so $N = S \times r = 2000 \times 3 = 6000 \text{ bps}$. L04 p5

29. MULTIPLE CHOICE

In the slides' TCP/IP model, which layer corresponds to OSI layers 1 and 2?

- A. Internet
- B. Transport
- C. Application
- D. Network access ✓ correct

Why: Network access spans OSI 1–2; Internet = OSI 3; Transport = OSI 4; Application = OSI 5–7. L01 p3

30. MULTIPLE CHOICE

Compared with NRZ, the **minimum bandwidth** of Manchester coding is:

- A. Half as much
- B. The same
- C. One and a half times as much
- D. Twice as much ✓ correct

Why: Manchester sends two signal elements for every bit ($r = 1/2$), so its average signal rate is N while NRZ needs only $N/2$ — twice the bandwidth (slide 30; the table on slide 46 lists $B = N$ against $N/2$). L03 p30; L03 p46

Part II — Identification

20 questions × 1 pt = 20 pts

1. IDENTIFICATION

The third step of block coding, after division and substitution, in which the n -bit groups are joined back into one stream.

Answer: Combination (also accepted: combining, combine)

Why: Block coding is done in three steps: division, substitution and combination (slides 48–49). L03 p48

2. IDENTIFICATION

The scrambling technique that replaces eight consecutive zeros with 000VB0VB.

Answer: B8ZS (also accepted: bipolar with 8-zero substitution, b8zs scrambling)

Why: B8ZS: V = violation, B = bipolar pulse (slide 59). HDB3 is the four-zero counterpart (000V or B00V, slides 60–61). L03 p59

3. IDENTIFICATION

The data link layer service of partitioning data into frames with recognized frame boundaries.

Answer: Framing (also accepted: frame)

Why: Framing partitions data into frames and exchanges them over the link. L01 p5

4. IDENTIFICATION

The block code whose encoder splits each 8-bit block between a 5B/6B encoder and a 3B/4B encoder and passes the result through a disparity controller.

Answer: 8B/10B (also accepted: 8B/10B encoding, 8B/10B block coding, 8B/10B block encoding, 8B/10B block code, 8B/10B code)

Why: Slide 55: 8-bit block → 5B/6B + 3B/4B encoding → disparity controller → 10-bit block. Slide 56: the extra redundancy lets it choose code words that avoid long runs of one level (DC components). L03 p55; L03 p56

5. IDENTIFICATION

The sampling method that holds each sample at one constant amplitude for the pulse duration.

Answer: Flat-top sampling (also accepted: flat top sampling, flat-top, flat top, sample and hold)

Why: Flat-top sampling is "sample and hold": like natural sampling, but each pulse keeps a single amplitude. L03 p64

6. IDENTIFICATION

The polar scheme that uses three voltage values (+, 0, -) and has a transition in the middle of every symbol, from high or low to zero.

Answer: RZ (also accepted: Return to Zero, Return-to-Zero, polar RZ, RZ scheme, RZ coding, return to zero scheme, return to zero coding)

Why: RZ (Return to Zero): a 1 is +V then 0 V, a 0 is -V then 0 V. It has two transitions per symbol, so it needs a wider bandwidth, but no DC component and it self-synchronizes (slides 26–27). L03 p26

7. IDENTIFICATION

The form of binary ASK in which a 1 sends the carrier and a 0 sends no signal at all.

Answer: On-off keying (also accepted: OOK, on off keying, binary ASK, BASK)

Why: Binary ASK where the amplitude for one bit value is zero is called on-off keying (OOK). L04 p8; L04 p10

8. IDENTIFICATION

A measure of how fast we can actually send data through a network (always less than the link's bandwidth).

Answer: Throughput

Why: A link may have a bandwidth of B bps, but we can only send T bps through it, with T always less than B. L02 p34

9. IDENTIFICATION

The loss of some of a signal's energy as it travels through a medium, in overcoming the resistance of the medium.

Answer: Attenuation (also accepted: signal attenuation)

Why: Attenuation means a loss of energy; amplifiers are used to compensate for it, and the change in strength is measured in decibels. L02 p20

10. IDENTIFICATION

The kind of noise in which signals jump from one wire to another, normally between adjacent wires.

Answer: Crosstalk (also accepted: cross talk)

Why: Crosstalk is signals jumping from one wire to another. L02 p24

11. IDENTIFICATION

A change in the form or shape of a composite signal because its frequency components propagate at different speeds.

Answer: Distortion (also accepted: signal distortion)

Why: Distortion means the signal changes its form or shape; the components of a composite signal arrive at different times. L02 p23

12. IDENTIFICATION

The digital-to-analog scheme that changes both the amplitude and the phase of the carrier.

Answer: QAM (also accepted: quadrature amplitude modulation)

Why: QAM is a combination of ASK and PSK. L04 p30

13. IDENTIFICATION

The diagram that shows each signal element as a point whose distance from the origin is its amplitude and whose angle is its phase.

Answer: Constellation diagram (also accepted: constellation)

Why: A constellation diagram uses the in-phase carrier as the x-axis and the quadrature carrier as the y-axis. L04 p27; L04 p28

14. IDENTIFICATION

The technique that compresses sample values at the sender and expands them at the receiver to keep the SNQR more even across amplitudes.

Answer: Companding (also accepted: compression and expansion, compand)

Why: Companding = compressing (logarithmic zones) then expanding; it helps low-amplitude signals that suffer most from quantization error. L03 p78

15. IDENTIFICATION

The kind of noise caused by the random motion of electrons in a wire.

Answer: Thermal noise (also accepted: thermal)

Why: Thermal noise is the random motion of electrons in a wire. Induced noise comes from motors and appliances, crosstalk from neighbouring wires, impulse noise from power lines and lightning. L02 p24

16. IDENTIFICATION

In the 4B/5B table, the control sequence **11111** is named:

Answer: Idle (also accepted: I, I (Idle), Idle (I))

Why: Slide 51 control sequences: Q (Quiet) 00000, I (Idle) 11111, H (Halt) 00100, J and K (Start delimiters) 11000 and 10001, T (End delimiter) 01101, S (Set) 11001, R (Reset) 00111. L03 p51

17. IDENTIFICATION

The coverage unit a wireless network uses to interconnect mobile devices.

Answer: Cell (also accepted: cells)

Why: Wireless networks use a "cell" to interconnect mobile devices. L01 p12

18. IDENTIFICATION

The OSI layer that translates frames into electrical, optical or electromagnetic signals representing 0s and 1s.

Answer: Physical layer (also accepted: physical, layer 1)

Why: The physical layer (layer 1) turns frames into signals and specifies cables, connectors and pin signals. L01 p14

19. IDENTIFICATION

The digitization scheme that sends a single bit per sample: 1 if the signal rose above the staircase, 0 if it did not.

Answer: Delta modulation (also accepted: DM)

Why: Delta modulation sends only the direction of change, one bit per sample. L03 p84

20. IDENTIFICATION

The term for unused fiber-optic cable.

Answer: Dark fiber (also accepted: dark fibre)

Why: Slide 10: "dark fiber" is used for unused fiber-optic cable. L01 p10

Part III — Draw: line encodings and waves

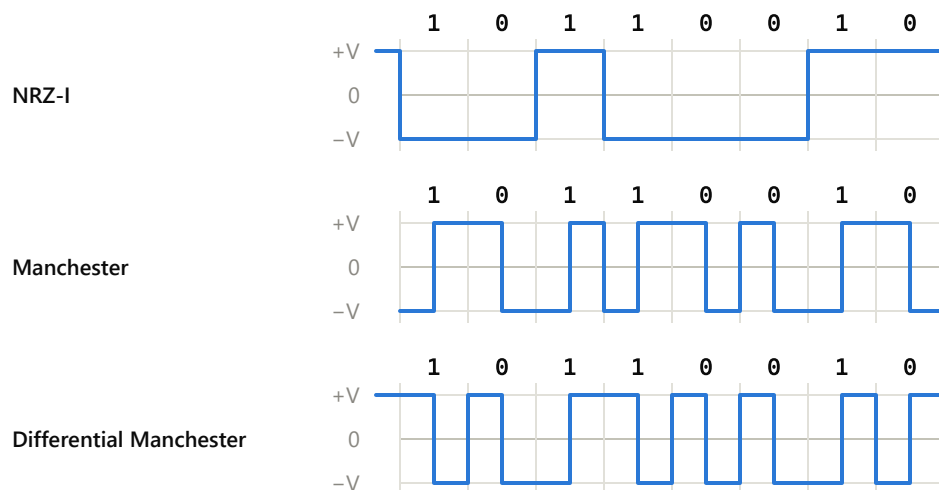
4 problems × 5 pts = 20 pts

1. PROBLEM · 5 PTS · L03A

Draw the bit stream **10110010** as a waveform in each of the line codes below (fill every cell of the grid). Use the slide conventions:

- NRZ-I: a 1 inverts the level at the start of the bit, a 0 does not; assume the level before the first bit is +V.
- Manchester: 0 = high→low, 1 = low→high (a transition in the middle of every bit).
- Differential Manchester: always a transition in the middle of the bit; a 0 also has a transition at the start of the bit, a 1 does not; assume the level before the first bit is +V.

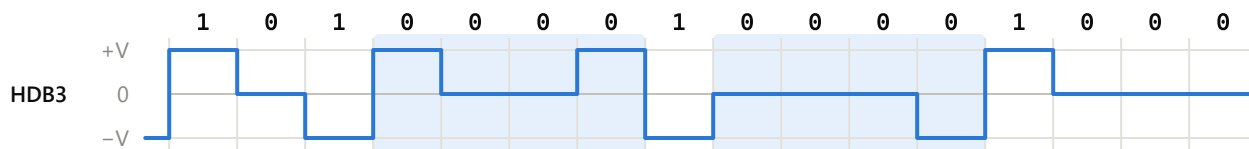
- NRZ-I** — a 1 inverts the level at the start of the bit, a 0 does not; assume the level before the first bit is +V. L03 p23
- Bit by bit: 1→-, 0→-, 1→+, 1→-, 0→-, 0→-, 1→+, 0→+ — the waveform is - - + - - - + +.
- Manchester** — 0 = high→low, 1 = low→high (a transition in the middle of every bit). L03 p29
- Bit by bit: 1→+, 0→+-, 1→++, 1→+-, 0→+-, 0→+-, 1→+-, 0→+- — the waveform is (-+)(+-)(-+)(-+)(+-)(+-)(-+)(+-).
- Differential Manchester** — always a transition in the middle of the bit; a 0 also has a transition at the start of the bit, a 1 does not; assume the level before the first bit is +V. L03 p29
- Bit by bit: 1→+-, 0→+-, 1→+-, 1→+-, 0→+-, 0→+-, 1→+-, 0→+- — the waveform is (+-)(+-)(-+)(+-)(+-)(+-)(-+)(-+).



2. PROBLEM · 5 PTS · L03A

Apply **HDB3** scrambling (on top of AMI) to the data stream **1010 0001 0000 1000** and draw the line signal. The last nonzero pulse sent before this stream was negative (-V). At the start of the stream the number of nonzero pulses since the last substitution is even.

- Scrambling works on top of AMI: 0 = 0 V and the 1s alternate in polarity. The last pulse before the stream was negative (-V), so the next 1 is +. L03 p58
- HDB3 replaces four consecutive 0s: count the nonzero pulses since the last substitution — **even** → **B00V**, **odd** → **000V** (the total becomes even either way). L03 p60
- Substitution 1 (bits 4–7): 2 nonzero pulses since the start → even → **B00V** = + 0 0 + (B alternates from the previous pulse, V repeats B). L03 p61
- Substitution 2 (bits 9–12): 1 nonzero pulse since the last substitution → odd → **000V** = 0 0 0 - (V repeats the previous pulse). L03 p61
- The transmitted pulses are + 0 - + 0 0 + - 0 0 0 - + 0 0 0 — no run of four zeros is left.



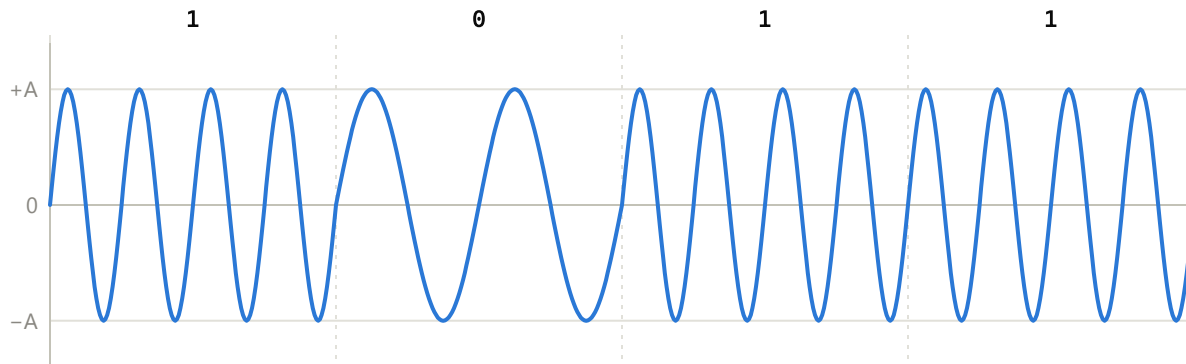
Pulse before the stream: -. Substitutions: bits 4–7 → B00V (even pulses since last substitution); bits 9–12 → 000V (odd pulses since last substitution). Marks: B at bit 4, V at bit 7, V at bit 12.

3. PROBLEM · 5 PTS · L04

Draw the **binary FSK** signal for the bits **1011**: a 1 uses the higher frequency $f_1 = f_c + \Delta f$, a 0 the lower frequency $f_2 = f_c - \Delta f$ (draw about 4 cycles per bit for f_1 and 2 for f_2).

1. The rule for binary FSK: a 1 uses the higher frequency $f_1 = f_c + \Delta f$, a 0 the lower frequency $f_2 = f_c - \Delta f$ (draw about 4 cycles per bit for f_1 and 2 for f_2). L04 p14
2. Bit by bit: 1 → f_1 (fast), 0 → f_2 (slow), 1 → f_1 (fast), 1 → f_1 (fast).
3. A binary scheme carries one bit per signal element ($r = 1$), so every bit cell has the same width.

Binary FSK for 1011



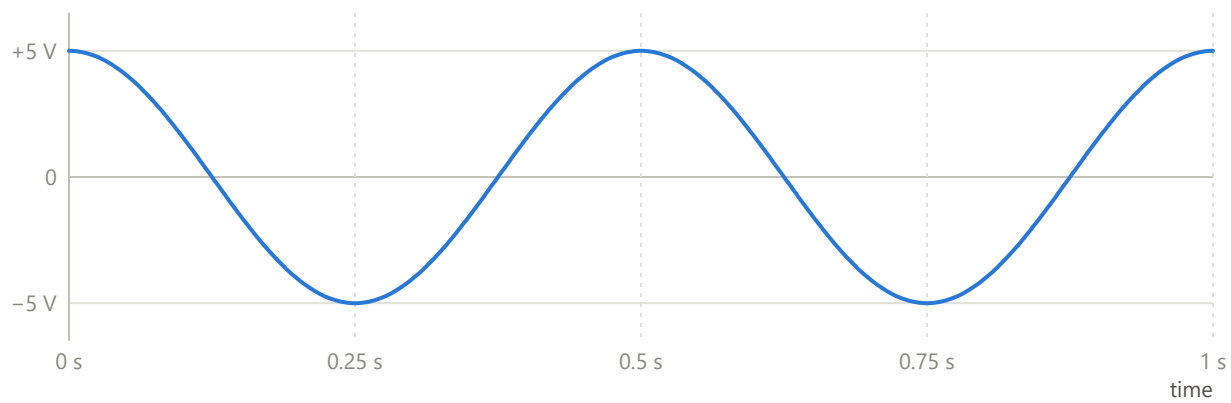
- Every 1-bit uses the higher frequency f_1 (more cycles per bit).
- Every 0-bit uses the lower frequency f_2 (fewer cycles per bit).
- The amplitude is the same in every bit.

4. PROBLEM · 5 PTS · L02

Draw one second of a sine wave with peak amplitude **5 V**, frequency **2 Hz** and phase **90°**.

1. Period: $T = 1/f = 1/2 \text{ Hz} = \mathbf{0.5 \text{ s}}$, so 2 cycles fit in 1 s. L02 p7
2. Phase 90°: the wave starts **at the peak**. L02 p8
3. Peak amplitude 5 V: the crests reach +5 V and the troughs -5 V. L02 p7

Sine wave, 0 to 1 s



- At $t = 0$ the wave starts at the peak (phase 90°).
 - 2 full cycles in 1 s — one cycle every $T = 1/f = 0.5$ s.
 - Crests at +5 V and troughs at -5 V.
-

Part IV — Problem solving (no calculator)

2 problems × 5 pts = 10 pts

1. PROBLEM · 5 PTS · L02

We have a channel with a **2 MHz** bandwidth. The SNR for this channel is **255**. Find the upper limit of the bit rate. For better performance we then use a lower bit rate of **12 Mbps**: how many signal levels are needed?

1. The channel is noisy, so Shannon gives the upper limit: $C = B \times \log_2(1 + \text{SNR}) = 2 \text{ MHz} \times \log_2(1 + 255) = 2 \text{ MHz} \times \log_2 256 = 2 \text{ MHz} \times 8 = \mathbf{16 \text{ Mbps}}$. L02 p31 L02 p33
2. We may send at most C, and a lower rate is safer: $N = 12 \text{ Mbps} \leq 16 \text{ Mbps}$. L02 p33
3. Nyquist links the rate to the levels: $N = 2 \times B \times \log_2 L$, so $\log_2 L = 12,000,000 / (2 \times 2,000,000) = 3$. L02 p29
4. $L = 2^3 = \mathbf{8}$ signal levels. L02 p33

Upper limit C = **16 Mbps** Signal levels L = **8**

2. PROBLEM · 5 PTS · L03B

A low-pass analog signal with frequencies up to **15 kHz** is digitized with PCM using **64 quantization levels**, sampled at the Nyquist rate. Find the sampling rate, the bits per sample, the bit rate, and the minimum bandwidth of the digitized signal (NRZ-type line code).

1. $f_{\max} = 15 \text{ kHz} \rightarrow$ Nyquist sampling rate $f_s = 2 \times f_{\max} = \mathbf{30 \text{ ksamples/s}}$. L03 p69
2. $n_b = \log_2 L = \log_2 64 = \mathbf{6 \text{ bits/sample}}$. L03 p75
3. Bit rate $N = n_b \times f_s = 6 \times 30 \text{ ksamples/s} = \mathbf{180 \text{ kbps}}$. L03 p79
4. Minimum bandwidth (NRZ-type line code, $c = 1/2$, $r = 1$): $B_{\min} = N/2 = n_b \times f_{\max} = \mathbf{90 \text{ kHz}}$ — versus only 15 kHz for the analog signal. L03 p83

Sampling rate $f_s = \mathbf{30 \text{ ksps}}$ Bits per sample $n_b = \mathbf{6}$ Bit rate $N = \mathbf{180 \text{ kbps}}$ Minimum bandwidth = **90 kHz**

Part V — Essay (draw and/or explain)

4 questions × 5 pts = 20 pts

1. ESSAY

Draw the **OSI** model and the slides' **TCP/IP** model side by side, showing which OSI layers each TCP/IP layer covers. Then explain the job of the two lowest OSI layers.

RUBRIC

- 1 pt** OSI, top to bottom: application, presentation, session, transport, network, data link, physical (7 layers).
- 1 pt** TCP/IP on the slides: application (= OSI application, presentation and session), transport, internet (= network), network access (= data link and physical).
- 1 pt** The drawing lines the two stacks up correctly ($3 + 1 + 1 + 2 = 7$).
- 1 pt** Physical layer: the physical structure — cables, connectors and how bit values become electrical, optical or electromagnetic signals.
- 1 pt** Data link layer: formats data into frames and moves them error-free between two end nodes over the physical layer (framing, frame sequencing, flow control, error detection, QoS).

MODEL ANSWER

OSI model (slide 2)

7	Application	End-user layer · HTTP, FTP, DNS
6	Presentation	Syntax layer · SSL, JPEG, MPEG
5	Session	Synch & send to port · sockets
4	Transport	End-to-end connections · TCP, UDP
3	Network	Packets · IP, ICMP
2	Data link	Frames · Ethernet, PPP, switch, bridge
1	Physical	Physical structure · coax, fiber, hubs

TCP/IP model (slide 3)

Application
Transport
Internet
Network access

The two bottom layers (highlighted) are the ones this deck reviews: the data link and physical layers.

OSI (7 layers)	TCP/IP on the slides (4 layers)
7 Application · 6 Presentation · 5 Session	Application
4 Transport	Transport
3 Network	Internet
2 Data link · 1 Physical	Network access

L01 p2 L01 p3 The **physical layer** is the physical structure: cable and connector types, the signal on each pin and how bit values become physical signals. L01 p14 The **data link layer** formats data into frames and moves them error-free between two end nodes over the physical layer; its services are framing, frame sequencing, flow control, error detection and QoS. L01 p4 L01 p5

L01 pp2–5; L01 p14

2. ESSAY

Explain how **PCM** turns an analog signal into a digital one. Draw a short analog signal and show what each of the three steps does to it, then explain why the sampling rate must be at least twice the highest frequency. You may draw, explain, or both.

RUBRIC

1 pt Sampling: the signal is measured every T_s ($f_s = 1/T_s$); the result is PAM — pulses that still have analog (non-integer) heights.

1 pt Quantization: the range is split into L zones of height $\Delta = (\text{max} - \text{min})/L$ and each sample is replaced by its zone midpoint; the error is at most $\Delta/2$.

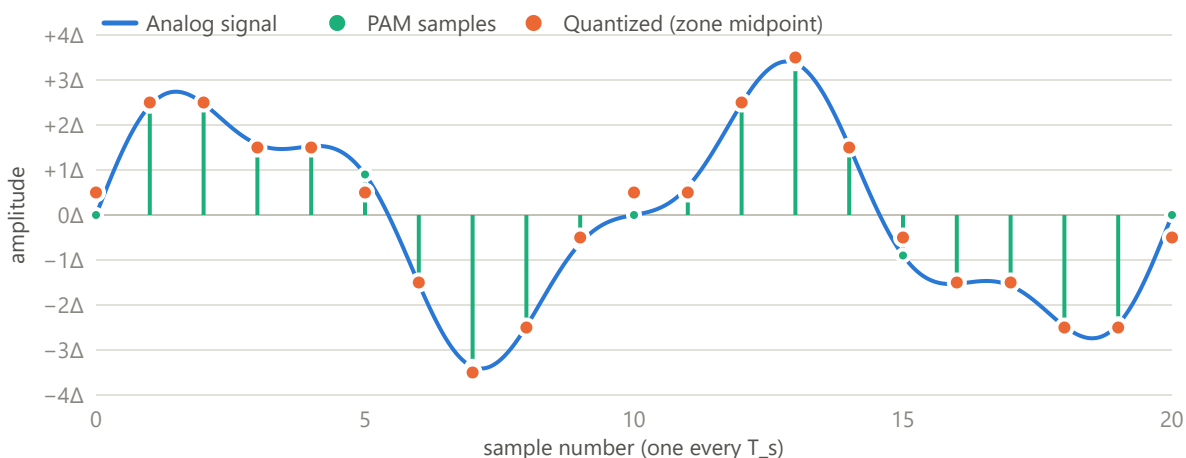
1 pt Encoding: each zone number is sent as n_b bits, $n_b = \log_2 L$, so the bit rate is $N = n_b \times f_s$.

1 pt Nyquist: $f_s \geq 2 f_{\text{max}}$; sampled more slowly, the samples trace a false, slower wave — so the signal is filtered first to limit f_{max} .

1 pt A labelled drawing (or an equally clear description) of the samples, the staircase of quantized values and the codes.

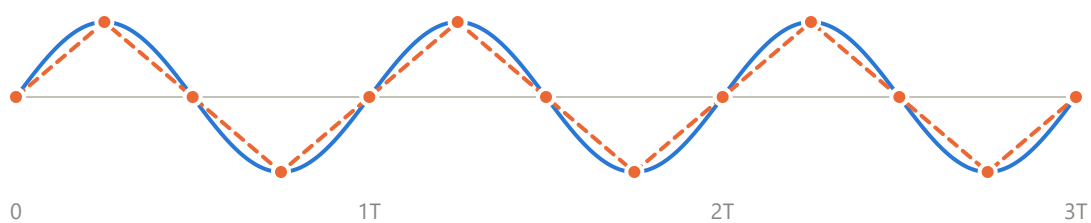
MODEL ANSWER

PCM has three steps. Sampling measures the analog signal every T_s seconds ($f_s = 1/T_s$); the result is a PAM signal — pulses whose heights are still analog values. [L03 p64](#) **Quantization** divides the range from min to max into L zones of height $\Delta = (\text{max} - \text{min})/L$ and replaces each sample by the midpoint of its zone; the difference is the quantization error, at most $\Delta/2$. [L03 p72](#) [L03 p78](#) **Encoding** sends each zone number as an n_b -bit code word, $n_b = \log_2 L$, so the bit rate is $N = n_b \times f_s$ (8 bits $\times$ 8000 samples/s = 64 kbps for voice). [L03 p75](#) [L03 p79](#) [L03 p69](#) [L03 p80](#)

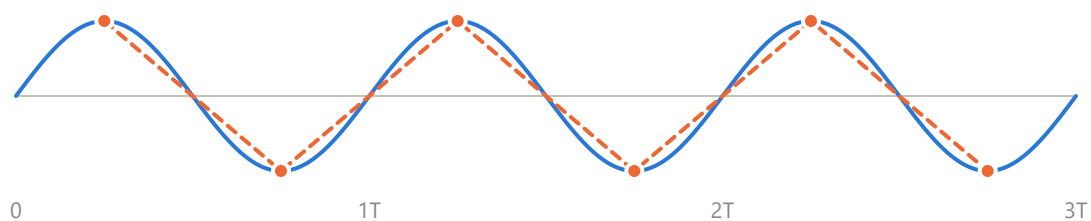


Why $2 \times f_{\text{max}}$: by the Nyquist theorem the sampling rate must be at least twice the highest frequency in the signal. Sampled more slowly, the samples trace a false, slower wave and the original cannot be rebuilt, so the signal is filtered first to limit f_{max} . [L03 p62](#) [L03 p66](#) [L03 p68](#)

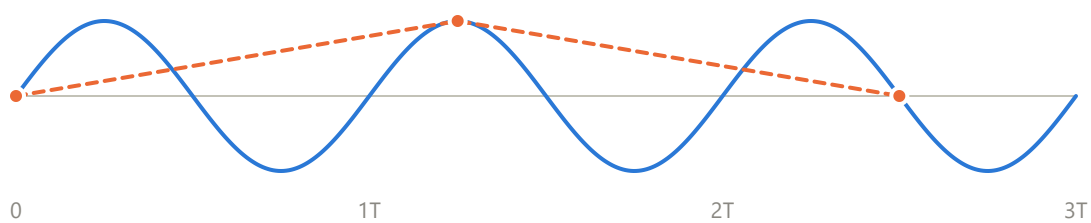
Oversampling: $f_s = 4f$ — the same shape, with redundant samples



Nyquist rate: $f_s = 2f$ — just enough to rebuild the wave



Undersampling: f_s below $2f$ — the samples trace a slower, false wave (the slide labels this $f_s = f$)



3. ESSAY

Compare **NRZ-L**, **NRZ-I**, **Manchester** and **AMI** for three properties: **baseline wandering**, **DC component** and **self-synchronization**. For each code, say whether it has the problem and explain *why*. You may draw a short bit stream (for example a 1 followed by a long run of 0s) in each code to support your answer.

RUBRIC

1 pt Defines the properties: baseline wandering = the receiver's running average (the baseline) drifts during long runs of one level; DC component = a constant level for long periods adds low-frequency content that bandpass channels may not pass; self-synchronization = the signal itself has enough transitions for the receiver's clock to stay in step with the sender's.

1 pt NRZ-L: a long run of 0s or 1s is a constant level, so it has baseline wandering and a DC component (worse than NRZ-I) and no self-synchronization.

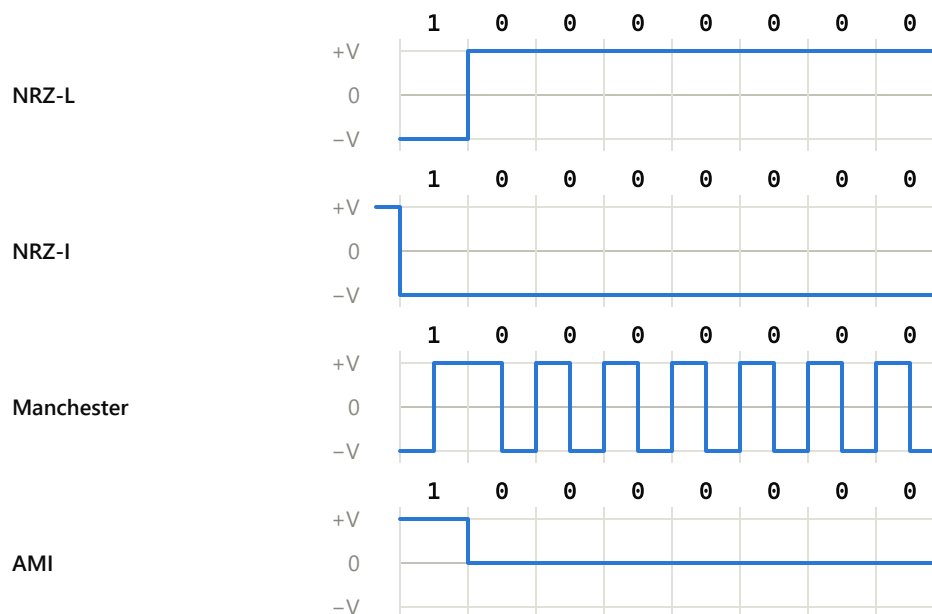
1 pt NRZ-I: a 1 always inverts the level, so runs of 1s keep transitions, but a long run of 0s is a constant level — DC component and baseline wandering remain (less than NRZ-L) and there is no self-synchronization for long 0s.

1 pt Manchester: a transition in the middle of every bit gives self-synchronization, and the average level is zero, so there is no DC component or baseline wandering — at the price of two signal elements per bit ($S = N$, twice the bandwidth of NRZ).

1 pt AMI: the 1s alternate in polarity, so there is no DC component or baseline wandering, but a long run of 0s (0 V) has no transitions, so no self-synchronization (the remedy is scrambling: B8ZS / HDB3).

MODEL ANSWER

All three properties come from **how long the signal stays at one level**.



A 1 then seven 0s: NRZ-L, NRZ-I and AMI stay flat through the 0s; Manchester still changes in the middle of every bit.

- **Baseline wandering** — the receiver decodes against a running average of the received signal power (the baseline). A long string of 0s or 1s drags that average away, so levels become hard to tell apart. [L03 p11](#)
- **DC component** — when the voltage stays constant for long periods the low frequencies of the signal grow; most channels are bandpass and may not pass them, so the DC component has to be avoided. [L03 p12](#)
- **Self-synchronization** — sender and receiver clocks must have the same bit interval; if the signal carries enough transitions, the receiver can re-align its clock to them. [L03 p13](#)

Code	Baseline wandering / DC	Self-synchronization	Why
NRZ-L	Yes — worse than NRZ-I	No	The level itself is the bit, so a long run of 0s or 1s is a flat line. L03 p24
NRZ-I	Yes — less severe	No (for long 0s)	A 1 inverts the level, so 1s keep producing transitions; a run of 0s is a flat line. L03 p24 L03 p46
Manchester	No	Yes	Every bit has a mid-bit transition (high→low or low→high), so the average is zero and the receiver sees a transition every bit. Cost: two signal elements per bit ($S = N$, twice NRZ's bandwidth). L03 p30
AMI	No	No (for long 0s)	The 1s alternate $+V$ / $-V$, so the positive and negative pulses cancel; but 0s are 0 V, and a long run of 0s has no transitions. L03 p33

Remedy for AMI: scrambling (B8ZS or HDB3) replaces long zero runs with violation pulses. L03 p57 Note: the summary table on slide 46 marks AMI "DC", which contradicts slide 33 — AMI has no DC component.

L03 pp11-13; L03 p24; L03 p30; L03 p33

4. ESSAY

Compare the **Nyquist** and **Shannon** formulas for the data-rate limit of a channel. For each, state the formula, when it applies and what it depends on. Then explain how the two are used together to choose a bit rate and a number of signal levels, using a channel with a 1 MHz bandwidth and an SNR of 63 as your example.

RUBRIC

- 1 pt** Nyquist: for a noiseless channel, bit rate = $2 \times \text{bandwidth} \times \log_2 L$ (L = number of signal levels).
- 1 pt** Shannon: for a noisy channel, capacity = $\text{bandwidth} \times \log_2(1 + \text{SNR})$, with the SNR as a plain ratio (not in dB).
- 1 pt** Dependencies: Nyquist depends on the bandwidth and the number of levels (more levels raise the rate, but the receiver must be able to tell them apart); Shannon depends on the bandwidth and the SNR only, so the number of levels does not dictate the capacity.
- 1 pt** Using both: Shannon gives the upper limit for a noisy channel; choose a lower bit rate for better performance, then solve Nyquist for the number of levels.
- 1 pt** Worked example (slide 33): $C = 10^6 \times \log_2(1 + 63) = 10^6 \times \log_2 64 = 6 \text{ Mbps}$; use 4 Mbps $\rightarrow 4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \rightarrow L = 4$.

MODEL ANSWER

Nyquist applies to a **noiseless** channel: *bit rate* = $2 \times \text{bandwidth} \times \log_2 L$, where L is the number of signal levels. It depends on the bandwidth and on the levels. Increasing the number of levels can increase the bit rate, but the receiver must be able to distinguish the levels.

Shannon applies to a **noisy** channel: *capacity* = $\text{bandwidth} \times \log_2(1 + \text{SNR})$, with the SNR as a plain ratio (an SNR given in dB must first be converted with $\text{SNR} = 10^{\text{SNR}_{\text{dB}}/10}$). It depends only on the bandwidth and the SNR: there is no signal-level term, so the number of levels does not dictate the capacity of the channel.

Used together: Shannon tells us the upper limit of the bit rate that the noisy channel can ever carry. We then pick a bit rate at or below that limit (a lower rate gives better performance) and use Nyquist, solved for L , to find how many signal levels that rate needs.

Example (slide 33): $B = 1 \text{ MHz}$ and $\text{SNR} = 63$. Shannon: $C = 10^6 \times \log_2(1 + 63) = 10^6 \times \log_2 64 = 6 \text{ Mbps}$, the upper limit. We choose 4 Mbps. Nyquist: $4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L$, so $\log_2 L = 2$ and $L = 4$ signal levels.

L02 pp28–33

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